TRANSPORT DEMAND PREDICTION

COURSE CODE: BCSE206L

COURSE TITLE: FOUNDATIONS OF DATA SCIENCE

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SLOT: F1+TF1

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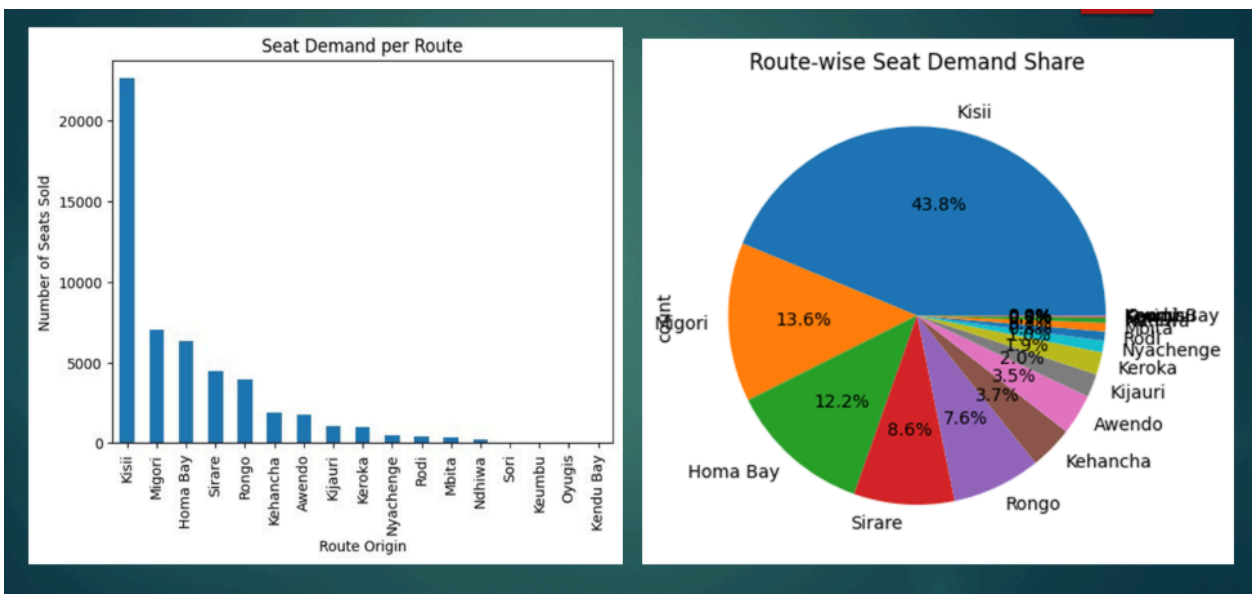
Project overview and objective:

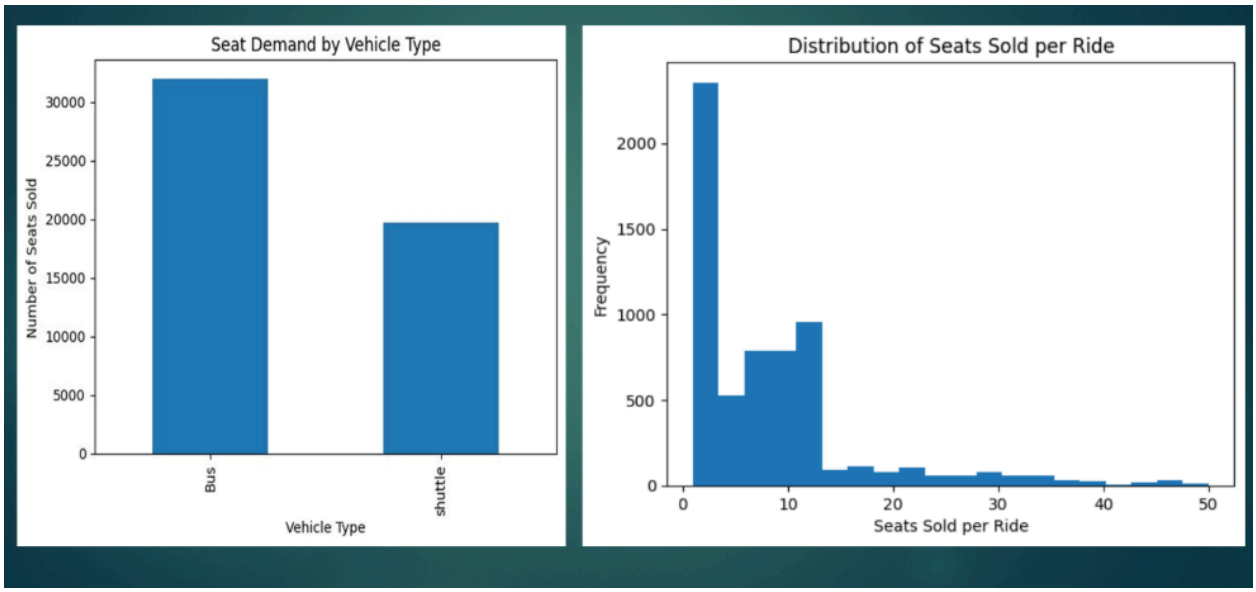
This project focuses on building a regression-based machine learning system to forecast the number of seats likely to be sold per ride. The goal is to help transportation providers better manage capacity, reduce seat wastage, and optimize pricing and scheduling decisions.

Objective:

- Better fleet and seat allocation
- Reduced overbooking and underutilization
- Improved operational efficiency
- Visualising the data in the form of graphs for better insights
- Data-driven decision-making for transport planning

Basic exploratory data analysis:





Workflow of the code:

The process starts by importing required libraries for data processing, modeling, evaluation, and visualization. It then loads the dataset 'train_final.csv' and groups it at the ride level using 'ride_id', converting raw passenger-level records into aggregated ride-level features such as passenger count, route details, timing, fare, demand behavior, and booking patterns.

Next, new engineered features are created like time format conversions, route/slot identifiers, interaction variables (e.g., fuel × distance), urgency metrics, and holiday/weekend combinations. Categorical variables like route and origin are encoded into numeric form using label encoding.

The dataset is then split into training and testing sets, followed by a more advanced step: K-Fold smoothed mean encoding, which replaces categorical values with statistically smoothed target means to improve predictive power while reducing overfitting.

After that, additional interaction features are generated by combining important variables (like demand × time, fare × distance, lead time × slot, etc.), and the final feature set is

prepared for modeling.

The code then trains a wide range of machine learning models including:

- Linear models (Linear Regression, Ridge, Lasso, ElasticNet, Huber)
- Distance-based models (KNN, SVR)
- Tree models (Decision Tree)
- Ensemble models (Random Forest, Bagging, Extra Trees)
- Boosting models (GBM, AdaBoost, XGBoost, LightGBM, CatBoost)

Each model is evaluated using R^2, MAE, and RMSE, and for boosting/ensemble models, performance is tracked across training iterations ("epochs" or trees) using staged predictions or warm-start checkpoints.

All predictions are stored and compared, and a final ranking of models is produced based on R^2 score.

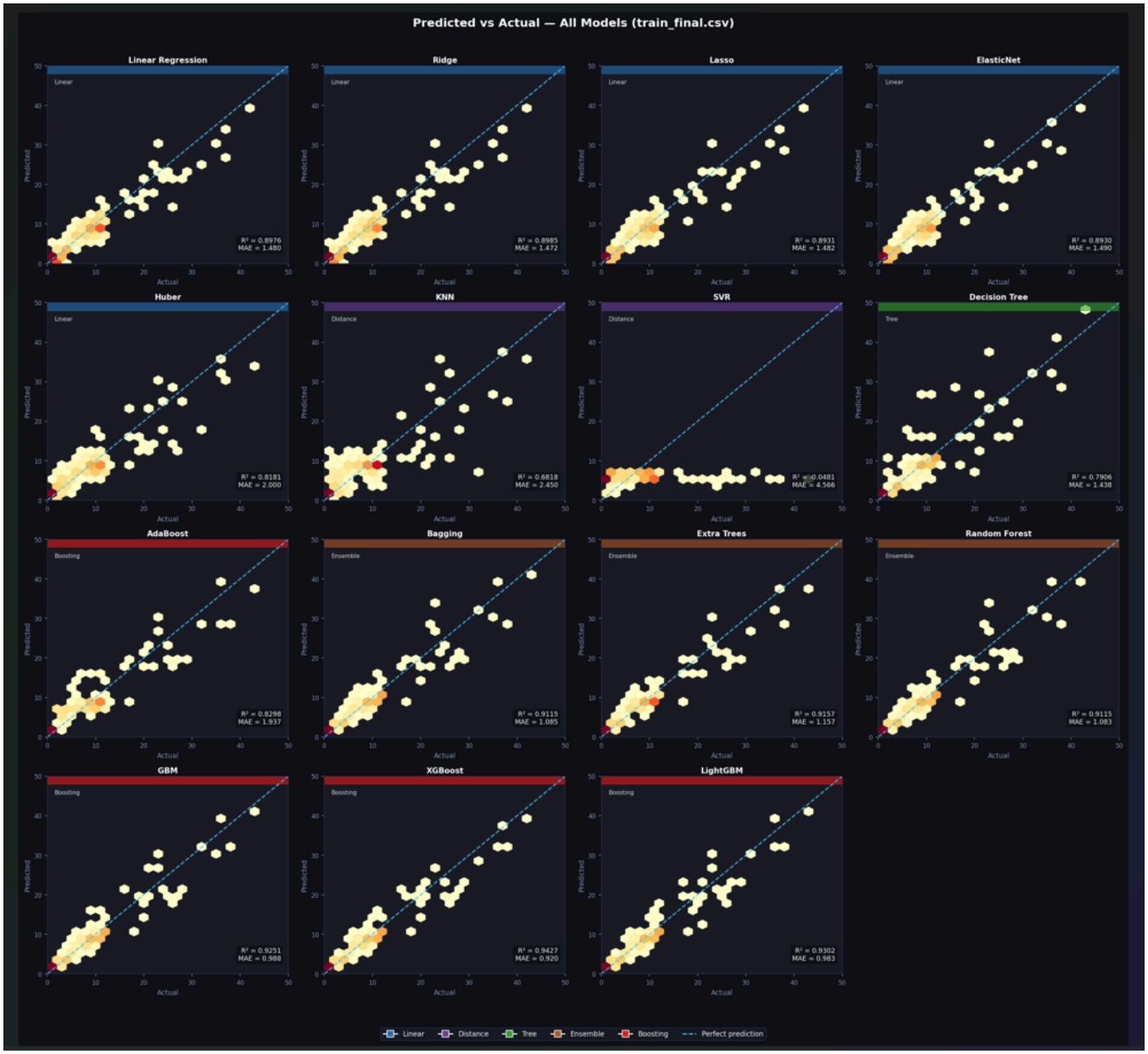
Finally, the code generates multiple visualizations:

- Predicted vs Actual plots for all models
- Model comparison charts (R^2, MAE, MSE, ranking)
- Learning curves for boosting/ensemble models

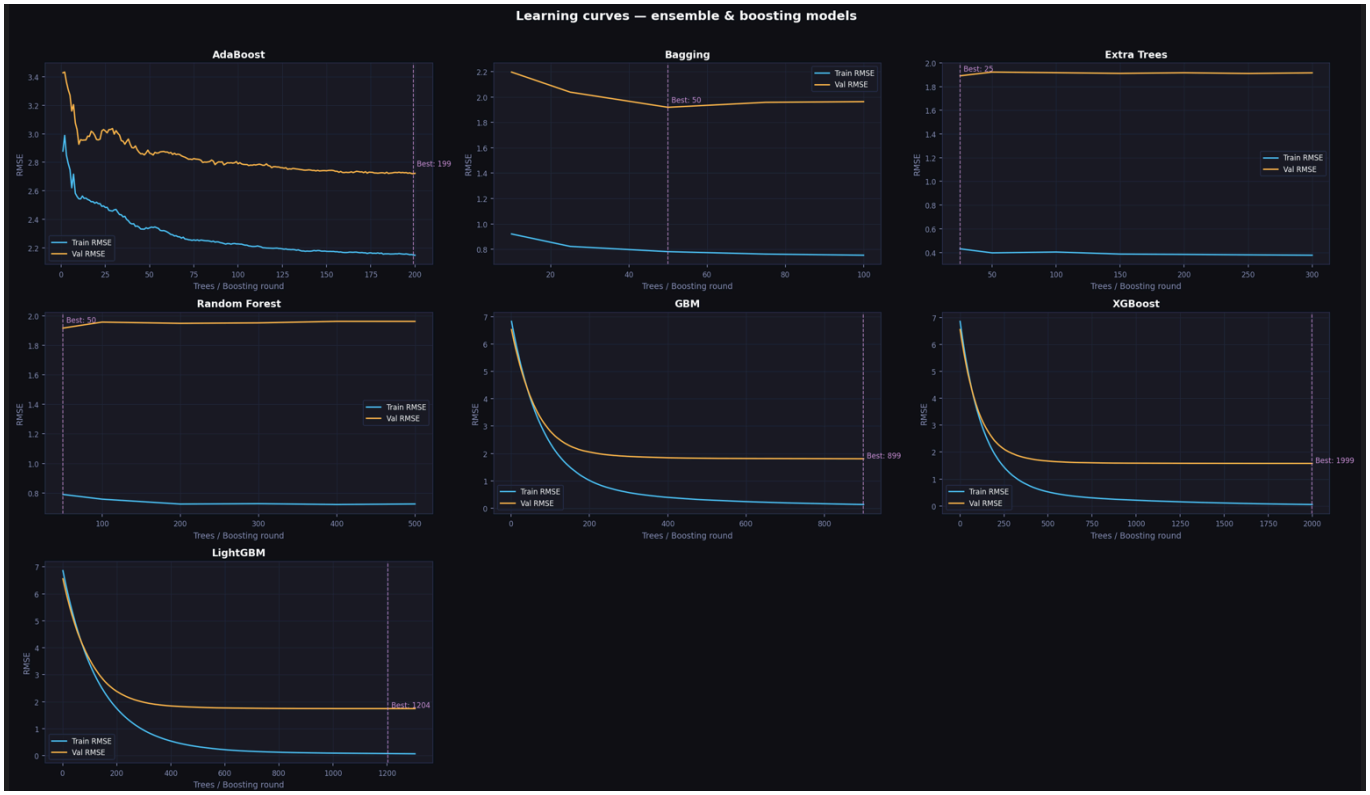
The feature importance section standardizes feature names, categorizes them (raw, booking, interaction, mean-encoded), computes importance scores, and plots them in a clean multi-model dashboard.

Screenshots of the outputs:

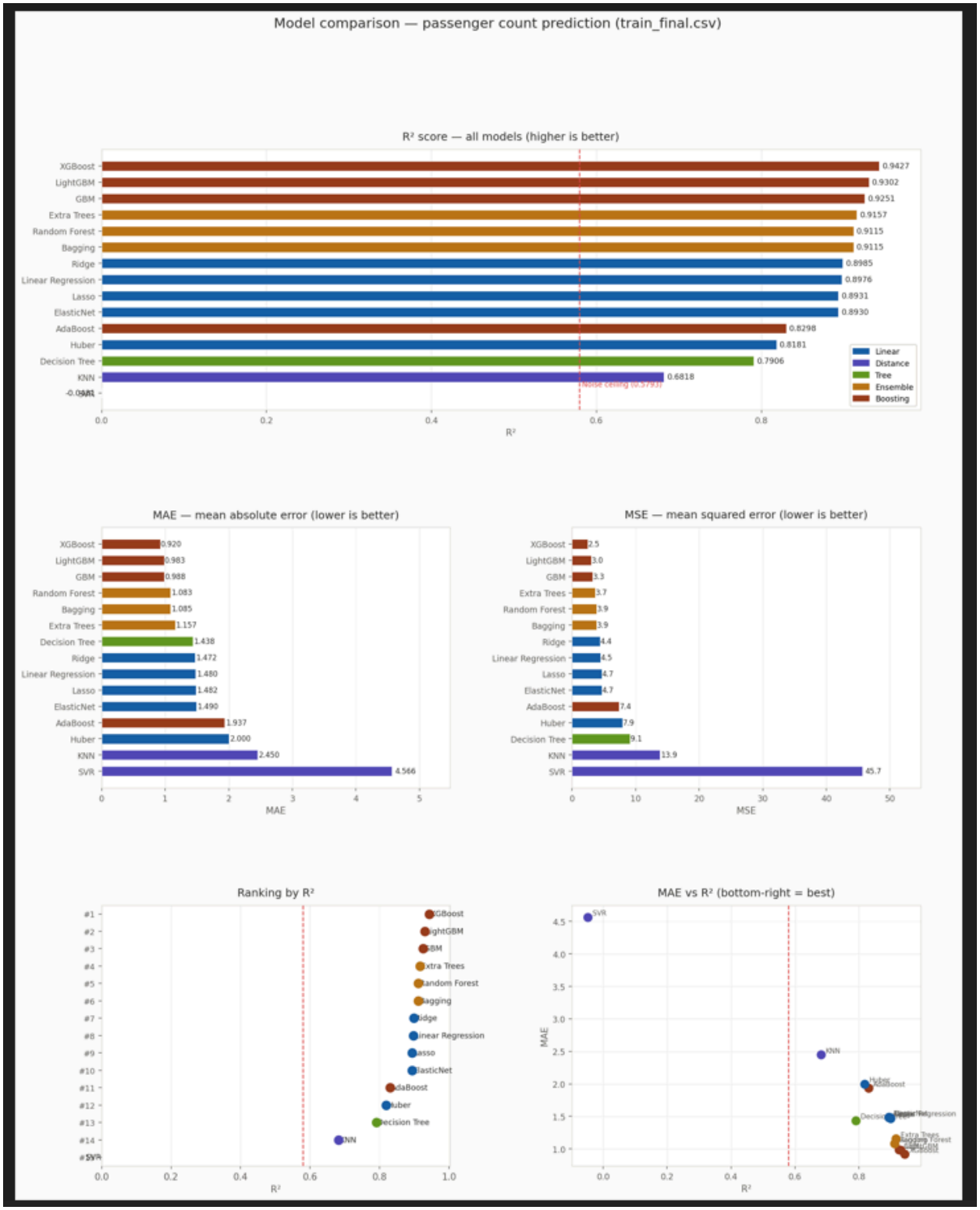
(i) Predicted vs Actual of every model:



(ii) Learning curves:



(iii) Model comparison:

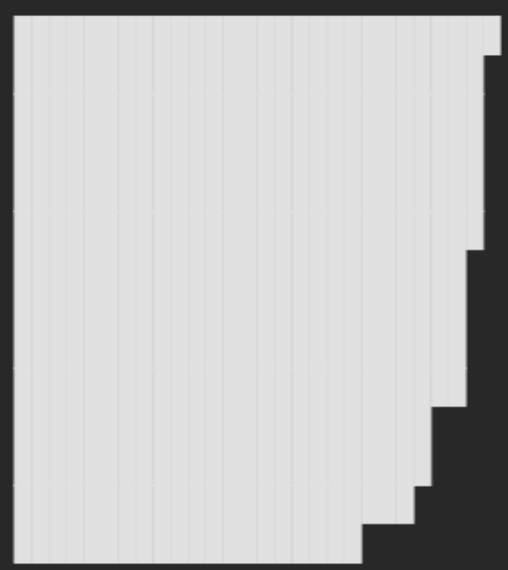


(iv) Evaluation metrics:

Model	R²	MAE	RMSE
Linear Regression	0.8976	1.480	2.113
Ridge	0.8985	1.472	2.103
Lasso	0.8931	1.482	2.159
ElasticNet	0.8930	1.490	2.159
Huber	0.8181	2.000	2.815
KNN	0.6818	2.450	3.723
SVR	-0.0481	4.566	6.758
Decision Tree	0.7906	1.438	3.021
AdaBoost	0.8298	1.937	2.723
Bagging	0.9115	1.085	1.964
Extra Trees	0.9157	1.157	1.917
Random Forest	0.9115	1.083	1.963
GBM	0.9251	0.988	1.806
XGBoost	0.9427	0.920	1.580
LightGBM	0.9302	0.983	1.745

--- Ranking by R² ---

1.	XGBoost	0.9427
2.	LightGBM	0.9302
3.	GBM	0.9251
4.	Extra Trees	0.9157
5.	Random Forest	0.9115
6.	Bagging	0.9115
7.	Ridge	0.8985
8.	Linear Regression	0.8976
9.	Lasso	0.8931
10.	ElasticNet	0.8930
11.	AdaBoost	0.8298
12.	Huber	0.8181
13.	Decision Tree	0.7906
14.	KNN	0.6818
15.	SVR	-0.0481



Key insights from this project:

1. Demand is highly time-dependent

Passenger demand varies significantly with hour, day of the week, and month. Peak hours such as morning and evening consistently show higher bookings, indicating strong commuter travel patterns.

2. Weekends and holidays significantly affect travel behavior

Features like ‘is_weekend’, ‘is_holiday’, and ‘is_school_hol’ show that demand changes noticeably during non-working days. Weekend travel is generally less predictable and often reflects leisure-driven patterns.

3. Route and origin are major demand drivers

Encoded features such as route, travel_from, and slot mean encoding are among the most important predictors. This shows that location-specific travel behavior plays a major role in demand variation.

4. Booking behavior strongly influences demand prediction

Variables such as booking urgency, ticket lead time, and lead spread significantly impact predictions. Shorter booking lead times are often associated with higher demand fluctuations and urgency-based travel.

5. Ensemble and boosting models perform best

Models like XGBoost, LightGBM, Random Forest, and CatBoost consistently outperform linear and distance-based models. This confirms that demand patterns are highly non-linear and involve complex feature interactions.

6. Feature engineering significantly improves performance

Interaction features such as fuel × distance, slot × week, and lead × holiday improve model accuracy. This shows that demand is influenced more by combined effects of variables rather than individual features.

7. Mean encoding captures hidden demand patterns

Smoothed target encoding features such as slot_mean, route_mean, and origin_mean help capture historical demand behavior effectively. These features contribute significantly to model performance improvements.

8. External factors also influence demand

Variables like fuel price, rainy season, distance, and competition levels affect travel demand, though their impact is smaller compared to time and route-based features.

9. Model agreement improves reliability

Top-performing models produce relatively consistent predictions, indicating stable learning patterns and strong underlying relationships in the dataset.

10. Overall conclusion

Passenger demand is best explained by a combination of time-based patterns, route characteristics, booking behavior, and seasonal effects. No single feature dominates; instead, demand is driven by interactions between multiple factors.

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